

MADUSHAN S H K

SOFTWARE ENGINEER — BACKEND / DISTRIBUTED SYSTEMS

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PROFESSIONAL SUMMARY

Backend Software Engineer with 4+ years of experience designing and operating production-grade microservices, event-driven architectures, and high-performance RESTful APIs in Go, Java, and Python. Proven record of improving system performance (30% reduction in API response times), enforcing data integrity at scale, and delivering cloud-native systems on AWS with Docker and fully automated CI/CD pipelines. Open-source contributor and author of a published Go concurrency-diagnostics library. Strong collaborator in remote, async, cross-functional teams; experienced mentoring junior engineers and driving architecture decisions.

CORE TECHNICAL SKILLS

Languages: Go (Golang), Java, Python, JavaScript

Frameworks & Libraries: Gin, Spring Boot, FastAPI, React.js, Vue.js / Vuetify

Architecture: Microservices, RESTful API Design, Event-Driven Architecture, Distributed Systems, System Design

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, GitHub Actions CI/CD, GitLab CI/CD

Databases & Messaging: MySQL, PostgreSQL, MongoDB (NoSQL), ZeroMQ (ZMQ)

Practices & Tools: Code Reviews, Unit & Integration Testing, SonarQube, Keycloak (Auth), Scrum, Agile

PROFESSIONAL EXPERIENCE

Software Engineer — Cerexio

Dec 2024 – Present

- Architected and developed the backend of a Manufacturing Execution System (MES) in Go (Gin) for an international client, supporting real-time manufacturing data at production scale.
- Optimized critical query paths and service logic, reducing average API response time by 30%.
- Designed RESTful microservices with strict data integrity constraints enforced at the MySQL layer.
- Authored comprehensive unit and integration test suites to safeguard production releases.
- Deployed and operated services on AWS (EC2, RDS, S3) using Docker containers with fully automated GitHub CI/CD pipelines.

Associate Software Engineer — VizuaMatix

Aug 2023 – Nov 2024

- Built and maintained RESTful APIs across a polyglot microservices architecture for a License Management System: Java/Spring Boot (payment service), FastAPI (user service), and Go (certificate generation service).
- Worked across the full backend stack, including PostgreSQL, MySQL, Keycloak for authentication, and SonarQube for code quality enforcement.
- Contributed to system architecture decisions and led code reviews to raise team-wide code quality.
- Mentored and supported junior engineers through onboarding and day-to-day development.

Trainee Software Engineer — VizuaMatix

Sep 2022 – Aug 2023

- Collaborated with senior developers to design and implement responsive user interfaces using the Vuetify framework, backed by Spring Boot services.

Software Engineer Intern — VizuaMatix

Feb 2022 – Sep 2022

- Designed, developed, and implemented RESTful APIs using Spring Boot, contributing to the backend infrastructure of production web applications.

OPEN SOURCE

gleak — Goroutine Leak Detector for Go (github.com/madushanshk98/gleak)

- Designed and published a zero-dependency Go library for detecting goroutine leaks in both production servers and test environments, addressing a gap left by existing tools that only work in tests and break under parallel execution.

SELECTED PROJECTS

License Management System (Government)

- Full-stack development of a government license management platform across a polyglot backend (Java/Spring Boot, FastAPI, Go) with Vue.js/Vuetify frontend and MySQL data layer.

EDUCATION

BSc (Hons) in Information Technology, Software Engineering — SLIIT

Jan 2020 – Dec 2023

Java Developer Certificate — Institute of Software Engineering (IJSE)

Dec 2019 – Apr 2020

CERTIFICATIONS

- Database Security for Cyber Professionals — May 2024
- AWS Services Fundamentals for Beginners — Simplilearn